

Implications of Theorem 8.3



- ▶ From the statement of Theorem 8.3 we can see that the number of mistakes m_T made by the WM algorithm is of the form $m_T \leq O(\log N) + \text{constant} * |\text{mistakes of best expert}|$.
- ▶ Since the first term varies only logarithmically with respect to the number of experts N , we see that the theorem guarantees that the number of mistakes made by the algorithm is roughly a constant times the number of mistakes made by the best expert in hindsight.
- ▶ This is a remarkable result in that it holds for any sequence of points and labels generated, including for sequences of points generated adversarially.

Limitations of the Weighted Majority algorithm



- ▶ In the previous lecture, we discussed the Weighted Majority algorithm for online learning.
- ▶ One key drawback of all deterministic algorithms for online learning in the case of zero-one loss is that the regret R_T cannot be $o(T)$ of all sequences.
- ▶ Recall from the order notation that $f(T)$ is $o(g(T))$ means $\lim_{T \rightarrow \infty} \frac{f(T)}{g(T)} = 0$.
- ▶ For any deterministic algorithm A and any $t \in [T]$ we can adversarially select y_t to be 1 if the algorithm selects 0 and 0 if the algorithm selects 1. Thus A errs at every point of the sequence and the cumulative number of mistakes over a sequence of length $m_T = T$.

Limitations of the Weighted Majority algorithm



- ▶ Assume that there are two experts so that $N = 2$. One of the experts always predicts 0 and the other always predicts 1.
- ▶ Let the sequence contain T points with k ones and $T - k$ zeroes. If $k < T/2$ then the expert who always predicts zeroes is the best expert and the error of the best expert $m_T^* < T/2$. On the other hand if $k \geq T/2$ then the expert who always predicts ones makes fewer than $T/2$ errors. Thus it is always the case that $m_T^* \leq T/2$.
- ▶ The regret for any sequence of length T is $R_T = m_T - m_T^* \geq T/2$.
- ▶ This shows that $R_T = o(T)$ cannot be achieved in general.



- ▶ We assume that a set $Q = \{1, 2, \dots, N\}$ of N actions is available.
- ▶ At each round $t \in [T]$ an online algorithm A selects a distribution $\mathbf{p}_t$ over the set of actions, receives a loss vector $\mathbf{l}_t$ whose i th component $l_{t,i} \in [0, 1]$ is the loss associated with action i , and incurs an expected loss $L_t = \sum_{i=1}^N p_{t,i} l_{t,i}$. The total loss incurred after T rounds is $\mathcal{L}_T = \sum_{t=1}^T L_t$.
- ▶ The total loss associated to action i after T rounds is $\mathcal{L}_{T,i} = \sum_{t=1}^T l_{t,i}$.
- ▶ The minimal loss of a single action is $\mathcal{L}_T^{\min} = \min_{i \in Q} \mathcal{L}_{T,i}$.



- ▶ The regret of the algorithm R_T after T rounds is defined by the difference of the loss of the algorithm and that of the best single action, i.e $R_T = \mathcal{L}_T - \mathcal{L}_T^{\min}$.
- ▶ We shall consider specifically the case of zero-one losses and assume that $l_{t,i} \in \{0,1\}$ for all $t \in [T]$ and $i \in Q$.
- ▶ We shall now present a straightforward generalization of the Weighted Majority algorithm called the Randomized Weighted Majority algorithm.

Randomized online learning



RANDOMIZED-WEIGHTED-MAJORITY (N)

```
1  for  $i \leftarrow 1$  to  $N$  do
2       $w_{1,i} \leftarrow 1$ 
3       $p_{1,i} \leftarrow 1/N$ 
4  for  $t \leftarrow 1$  to  $T$  do
5      RECEIVE( $I_t$ )
6      for  $i \leftarrow 1$  to  $N$  do
7          if  $(I_{t,i} = 1)$  then
8               $w_{t+1,i} \leftarrow \beta w_{t,i}$ 
9          else  $w_{t+1,i} \leftarrow w_{t,i}$ 
10          $W_{t+1} \leftarrow \sum_{i=1}^N w_{t+1,i}$ 
11         for  $i \leftarrow 1$  to  $N$  do
12              $p_{t+1,i} \leftarrow w_{t+1,i}/W_{t+1}$ 
13  return  $\mathbf{w}_{T+1}$ 
```

Theorem 8.4



Theorem 8.4 Fix $\beta \in [1/2, 1]$. Then, for any $T \geq 1$, the loss of algorithm RWM on any sequence can be bounded as follows:

$$\mathcal{L}_T \leq \frac{\log N}{1 - \beta} + (2 - \beta)\mathcal{L}_T^{\min}$$

In particular, for $\beta = \max\{1/2, 1 - \sqrt{\frac{\log N}{T}}\}$, the loss can be bounded as $\mathcal{L}_T \leq \mathcal{L}_T^{\min} + 2\sqrt{T \log N}$.

Proof of Theorem 8.4



- ▶ As in the proof of Theorem 8.3 we create a potential function $W_t = \sum_{i=1}^N w_{t,i}$, $t \in [T]$.
- ▶ We shall then find upper and lower bounds for this potential function.
- ▶ Following the working of the algorithm we can express the potential function at round $t + 1$, W_{t+1} in terms of W_t as on the next slide.

Proof of Theorem 8.4



$$\begin{aligned}W_{t+1} &= \sum_{i:l_t=0} w_{t,i} + \beta \sum_{i:l_t=1} w_{t,i} \\&= W_t + (\beta - 1) \sum_{i:l_t=1} w_{t,i} \\&= W_t + (\beta - 1) W_t \sum_{i:l_t=1} p_{t,i} \\&= W_t + (\beta - 1) W_t L_t \\&= W_t (1 - (1 - \beta) L_t).\end{aligned}$$

Proof of Theorem 8.4



- ▶ We see that $W_1 = N$. So $W_{T+1} = N \prod_{t=1}^{t=T} (1 - (1 - \beta)L_t)$.
- ▶ On the other hand the following lower-bound clearly holds:
 $W_{T+1} \geq \max_{i \in [N]} w_{T+1,i} = \beta^{\mathcal{L}_T^{\min}}$.
- ▶ This leads to the following inequality

$$\beta^{\mathcal{L}_T^{\min}} \leq N \prod_{t=1}^{t=T} (1 - (1 - \beta)L_t)$$

- ▶ We can now perform a series of simplifications on this inequality, including taking log, using $\log(1 - x) \leq -x, \forall x < 1$, and $-\log(1 - x) \leq x + x^2, \forall x \in [0, 1/2]$.

Proof of Theorem 8.4



$$\mathcal{L}_T^{\min} \log \beta \leq \log N + \sum_{t=1}^{t=T} \log(1 - (1 - \beta)L_t)$$

$$\Rightarrow \mathcal{L}_T^{\min} \log \beta \leq \log N - (1 - \beta) \sum_{t=1}^{t=T} L_t$$

$$\Rightarrow \mathcal{L}_T^{\min} \log \beta \leq \log N - (1 - \beta)\mathcal{L}_T$$

$$\Rightarrow \mathcal{L}_T \leq \frac{\log N}{1 - \beta} - \frac{\log \beta}{1 - \beta} \mathcal{L}_T^{\min}$$

$$\Rightarrow \mathcal{L}_T \leq \frac{\log N}{1 - \beta} - \frac{\log(1 - (1 - \beta))}{1 - \beta} \mathcal{L}_T^{\min}$$

$$\Rightarrow \mathcal{L}_T \leq \frac{\log N}{1 - \beta} + (2 - \beta)\mathcal{L}_T^{\min}$$

Proof of Theorem 8.4



- ▶ Since $\mathcal{L}_T^{\min} \leq T$ we can rewrite the last expression on the previous slide as

$$\mathcal{L}_T \leq \frac{\log N}{1-\beta} + (1-\beta)T + \mathcal{L}_T^{\min}$$

- ▶ Differentiating the upper bound with respect to β and setting it to zero gives $\frac{\log N}{(1-\beta)^2} - T = 0$, that is $\beta = 1 - \sqrt{\frac{\log N}{T}} < 1$.
- ▶ If $1 - \sqrt{\frac{\log N}{T}} > 1/2$, then $\beta_0 = 1 - \sqrt{\frac{\log N}{T}}$ is the minimizing value of β . Otherwise $\beta_0 = 1/2$ is the optimum value for β .
- ▶ Using $\beta = \beta_0$ in $\frac{\log N}{1-\beta} + (1-\beta)T + \mathcal{L}_T^{\min}$ gives us $\mathcal{L}_T \leq \mathcal{L}_T^{\min} + 2\sqrt{T \log N}$. Note that this result is valid for $T > 4 \log N$ since that will ensure that $1 - \sqrt{\frac{\log N}{T}} > \frac{1}{2}$.

Implications of Theorem 8.4



- ▶ As a consequence of Theorem 8.4 we can write $\mathcal{L}_T \leq \mathcal{L}_T^{\min} + 2\sqrt{T \log N}$. Since the regret $R_T = \mathcal{L}_T - \mathcal{L}_T^{\min}$, we can write $R_T = 2\sqrt{T \log N}$.
- ▶ When the number of experts N is a constant, we have $R_T = O(\sqrt{T})$ which could not be achieved by the WM algorithm.
- ▶ We now show that this bound is optimal.

Theorem 8.5



Theorem 8.5 Let $N = 2$. There exists a stochastic sequence of losses for which the regret of any online learning algorithm verifies

$$E[R_T] \geq \sqrt{\frac{T}{8}}.$$

Proof of Theorem 8.5



- ▶ For any $t \in [T]$ let the vector of losses $\mathbf{l}_t$ take the values $\mathbf{l}_{01} = (0, 1)^T$ and $\mathbf{l}_{10} = (1, 0)^T$ with equal probability.
- ▶ The expected loss of any randomized algorithm is A is

$$\begin{aligned}\mathbb{E}[\mathcal{L}_T] &= \mathbb{E}\left[\sum_{t=1}^{t=T} \mathbf{p}_t \cdot \mathbf{l}_t\right] \\ &= \sum_{t=1}^{t=T} \mathbf{p}_t \cdot \mathbb{E}[\mathbf{l}_t] \\ &= \sum_{t=1}^{t=T} \frac{1}{2} p_{t,1} + \frac{1}{2} (1 - p_{t,1}) = \frac{T}{2}\end{aligned}$$

Proof of Theorem 8.5



- ▶ Note that in the equation on the previous slide $\mathbf{p}_t$ is a probability distribution over the choice of experts whereas $\mathbf{I}_t$ is a random vector denoting what advice each of the experts chooses at time t . Note that in the previous slide we computed $E[\mathbf{I}_t] = \frac{1}{2}[0, 1] + \frac{1}{2}[1, 0] = [1/2, 1/2]$.
- ▶ We can write $\mathcal{L}_T^{\min}$ in the following way:

$$\begin{aligned}\mathcal{L}_T^{\min} &= \min(\mathcal{L}_{T,1}, \mathcal{L}_{T,2}) \\ &= \frac{1}{2}(\mathcal{L}_{T,1} + \mathcal{L}_{T,2} - |\mathcal{L}_{T,1} - \mathcal{L}_{T,2}|) \\ &= \frac{T}{2} - |\mathcal{L}_{T,1} - \frac{T}{2}|\end{aligned}$$

- ▶ Note that in the last step we have made use of the following equality: $T = \mathcal{L}_{T,1} + \mathcal{L}_{T,2}$.

Proof of Theorem 8.5



- ▶ We can therefore see that the expected regret $\mathbb{E}[R_T] = \mathbb{E}[\mathcal{L}_T] - \mathbb{E}[\mathcal{L}_T^{\min}] = \mathbb{E}[|\mathcal{L}_{T,1} - \frac{T}{2}|]$.
- ▶ Now let $\sigma_t, t \in [T]$ denote variables that take values in $\{-1, 1\}$
- ▶ Now we can write $\mathcal{L}_{T,1} = \sum_{t=1}^{t=T} \frac{1+\sigma_t}{2}$ since $\mathcal{L}_{T,1}$ is either 0 or 1.
- ▶ We can write $\mathcal{L}_{T,1} = \frac{T}{2} + \frac{1}{2} \sum_{t=1}^{t=T} \sigma_t$.

Proof of Theorem 8.5



- ▶ We now introduce scalars $x_t = \frac{1}{2}, t \in [T]$.
- ▶ By the Khintchine-Kahane inequality we have

$$\begin{aligned}\mathbb{E}[R_T] &= \mathbb{E}\left[\left|\sum_{t=1}^{t=T} \sigma_t x_t\right|\right] \\ &\geq \sqrt{\frac{1}{2} \sum_{t=1}^{t=T} x_t^2} \\ &\geq \sqrt{\frac{T}{8}}\end{aligned}$$

Exponential weighted average algorithm



- ▶ In the preceding lecture we discussed two online learning algorithms, the Weighted Majority (WM) algorithm and the Randomized Weighted Majority (RWM) algorithm
- ▶ The first of these algorithms suffers from a limitation that the regret of the algorithm is a constant times the number of rounds and is $O(T)$, not $o(T)$.
- ▶ In contrast the RWM algorithm exhibits a regret whose dependence on T the number of rounds is $O(\sqrt{T})$ which is an attractive feature since it is $o(T)$.
- ▶ Can we come up with a deterministic algorithm that exhibits the same dependence on T as the RWM algorithm?

Exponential weighted average algorithm



- ▶ We now explore a deterministic algorithm which has a very favourable regret guarantee and can be extended to loss functions L that are convex in their first argument.
- ▶ It is very similar to the WM algorithm in that in each round the algorithm makes a prediction $\hat{y}_t$ which is then compared to the true value y_t .
- ▶ A loss is computed as a function of $\hat{y}_t$ and y_t , and the weights are updated depending on the loss.
- ▶ The pseudo-code for this algorithm is shown on the next slide.

Exponential weighted average algorithm



EXPONENTIAL-WEIGHTED-AVERAGE (N)

```
1  for  $i \leftarrow 1$  to  $N$  do
2       $w_{1,i} \leftarrow 1$ 
3  for  $t \leftarrow 1$  to  $T$  do
4      RECEIVE( $x_t$ )
5       $\hat{y}_t \leftarrow \frac{\sum_{i=1}^N w_{t,i} y_{t,i}}{\sum_{i=1}^N w_{t,i}}$ 
6      RECEIVE( $y_t$ )
7      for  $i \leftarrow 1$  to  $N$  do
8           $w_{t+1,i} \leftarrow w_{t,i} e^{-\eta L(\hat{y}_{t,i}, y_t)}$ 
9  return  $\mathbf{w}_{T+1}$ 
```

Exponential weighted average algorithm



- ▶ At round $t \in [T]$, the algorithm's prediction is $\frac{\sum_{i=1}^N w_{t,i} \hat{y}_{t,i}}{\sum_{i=1}^N w_{t,i}}$.
Here $\hat{y}_{t,i}$ is the prediction of the i th expert at round t and $w_{t,i}$ is the weight assigned to the i th expert.
- ▶ Initially all weights are set to 1.
- ▶ The algorithm then updates the weights at the end of round t according to the rule $w_{t+1,i} \leftarrow w_{t,i} e^{-\eta L(\hat{y}_{t,i}, y_t)}$.
- ▶ Since $L(t, i)$ is the total loss incurred by the i th expert after t rounds, we can see that $w_{t+1,i} = e^{-\eta L(t,i)}$.
- ▶ Thus in order to determine the weight of each expert after t rounds, we do not need to maintain the actual history of each loss; it suffices to have only the cumulative performance of each expert.

Theorem 8.6



Theorem 8.6 Assume that the loss function L is convex in its first argument, and takes values in $[0, 1]$. Then for any $\eta > 0$ and $y_1, y_2, \dots, y_T \in \mathcal{Y}$, the regret of the Exponential Weighted Algorithm satisfies $R_T \leq \frac{\log N}{\eta} + \frac{\eta T}{8}$. In particular for $\eta = \sqrt{\frac{8 \log N}{T}}$, the regret is bounded as follows: $R_T \leq \sqrt{\frac{T}{2} \log N}$.

Proof of Theorem 8.6



- ▶ We apply the same potential function analysis as for the previous theorems establishing regret bounds, but our potential function is $\phi_t = \log \sum_{i=1}^N w_{t,i}$.
- ▶ Let $\mathbf{p}_t$ denote the distribution over $\{1, 2, \dots, N\}$ with $p_{t,i} = \frac{w_{t,i}}{\sum_{i=1}^N w_{t,i}}$.
- ▶ We first derive an upper bound on ϕ_t .
- ▶ To this end let us take the difference between two consecutive values of ϕ_t .

- ▶ Thus we have $\phi_{t+1} - \phi_t = \log \frac{\sum_{i=1}^N w_{t,i} e^{-\eta L(\hat{y}_{t,i}, y_t)}}{\sum_{i=1}^N w_{t,i}} = \log \mathbb{E}_{\mathbf{p}_t}[e^{\eta X}]$
with $X = -L(\hat{y}_{t,i}, y_t) \in [-1, 0]$.

Proof of Theorem 8.6



- ▶ We now apply Hoeffding's lemma to the right hand side of the expression on the previous slide.
- ▶ Hoeffding's lemma can be stated as follows: for a random variable X such that $E(X) = 0$ and $a \leq X \leq b$
$$\mathbb{E}[e^{tX}] \leq e^{\frac{t^2(b-a)^2}{8}}.$$
- ▶ In the calculations that follow on the next two slides, we use both Hoeffding's lemma as well as the implications of convexity of the function L on the first argument.

Proof of Theorem 8.6



$$\begin{aligned}\phi_{t+1} - \phi_t &= \log \mathbb{E}_{\mathbf{p}_t}[e^{\eta X}] \\&= \log \mathbb{E}_{\mathbf{p}_t}[e^{\eta(X-E(X)) + \eta E(X)}] \\&= \log e^{\eta E(X)} \mathbb{E}_{\mathbf{p}_t}[e^{\eta(X-E(X))}] \\&= \eta \mathbb{E}_{\mathbf{p}_t}[X] + \log \mathbb{E}_{\mathbf{p}_t}[e^{\eta(X-E(X))}] \\&= \eta \mathbb{E}_{\mathbf{p}_t}[X] + \log e^{\frac{\eta^2((0-q)-(-1-q))^2}{8}}, \text{ where } q = E(X) \\&= \frac{\eta^2}{8} + \eta \mathbb{E}_{\mathbf{p}_t}[X] = \frac{\eta^2}{8} - \eta \mathbb{E}_{\mathbf{p}_t}[L(\hat{y}_{t,i}, y_t)]\end{aligned}$$

Proof of Theorem 8.6



$$\begin{aligned}\phi_{t+1} - \phi_t &= \frac{\eta^2}{8} - \eta \mathbb{E}_{\mathbf{p}_t}[L(\hat{y}_{t,i}, y_t)] \\ &\leq \frac{\eta^2}{8} - \eta L(\mathbb{E}[\hat{y}_{t,i}], y_t) \\ &= \frac{\eta^2}{8} - \eta L(\hat{y}_t, y_t)\end{aligned}$$

Proof of Theorem 8.6



- ▶ Summing up the left hand sides of all the inequalities for different values of t yields a telescoping sum, so that
$$\phi_{T+1} - \phi_1 \leq -\eta \sum_{t=1}^T L(\hat{y}_t, y_t) + \frac{\eta^2 T}{8}.$$
- ▶ Having obtained an upper bound for $\phi_{T+1} - \phi_1$, we now develop a lower bound for this quantity.
- ▶ We note that after T rounds, the i th weight becomes $e^{-\eta L_{T,i}}$ where $L_{T,i}$ is the loss associated with the i th action after T rounds.

Proof of Theorem 8.6



Thus we have

$$\begin{aligned}\phi_{T+1} - \phi_1 &= \log \sum_{i=1}^{i=N} e^{-\eta L_{T,i}} - \log N \\ &\geq \log \max_{i=1}^{i=N} e^{-\eta L_{T,i}} - \log N \\ &= -\eta \min_{i=1}^{i=N} L_{T,i} - \log N\end{aligned}$$

Proof of Theorem 8.6



Combining the upper and lower bounds derived on the previous slides gives us

$$\begin{aligned} -\eta \min_{i=1}^{i=N} L_{T,i} - \log N &\leq \frac{\eta^2 T}{8} - \sum_{t=1}^{t=T} \eta L(\hat{y}_t, y_t) \\ \Rightarrow \sum_{t=1}^{t=T} L(\hat{y}_t, y_t) - \min_{i=1}^{i=N} L_{T,i} &\leq \frac{\log N}{\eta} + \frac{\eta T}{8} \end{aligned}$$

which completes the proof since $R_T = \sum_{t=1}^{t=T} L(\hat{y}_t, y_t) - \min_{i=1}^{i=N} L_{T,i}$.

Doubling trick



- ▶ The optimal choice of η in Theorem 8.6 requires prior knowledge of the value of T .
- ▶ We can use what is known as a doubling trick to eliminate this requirement and incur a price in the form of a constant factor.
- ▶ This consists of dividing the time interval into periods $[2^k, 2^{k+1} - 1]$ of length 2^k for $k = 0, 1, \dots, n$ and $T \geq 2^n - 1$.
- ▶ We can choose $\eta_k = \sqrt{\frac{8 \log N}{2^k}}$ in each interval.
- ▶ The theorem stated on the next slide presents a regret bound when using the aforementioned trick to select η .

Theorem 8.7



Theorem 8.7 Assume that the loss function L is convex in its first argument and takes values in $[0, 1]$. Then for any $T \geq 1$ and any sequence $y_1, y_2, \dots, y_T \in \mathcal{Y}$, the regret of the Exponential Weighted Average algorithm after T rounds is bounded as follows:

$$R_T \leq \frac{\sqrt{2}}{\sqrt{2}-1} \sqrt{\frac{T}{2} \log N} + \sqrt{\frac{\log N}{2}}$$

Proof of Theorem 8.7



- ▶ Let $T \geq 1$ and let $\mathcal{J}_k = [2^k, 2^{k+1} - 1]$ for $k \in [0, n]$ with $n = \lfloor \log_2(T + 1) \rfloor$.
- ▶ Let $L_{\mathcal{J}_k}$ denote the loss in the interval $\mathcal{J}_k$.
- ▶ From the result of the previous theorem, we know that the regret in an interval T can be bounded as follows:

$$R_T = \sqrt{\frac{T}{2} \log N}.$$

- ▶ Applying this result in our context gives

$$L_{\mathcal{J}_k} - \min_{i=1}^{i=N} L_{\mathcal{J}_k, i} \leq \sqrt{\frac{2^k}{2} \log N}.$$

Proof of Theorem 8.7



We can now bound the total loss after T rounds as follows:

$$\begin{aligned} L_T &= \sum_{k=0}^{k=N} L_{\mathcal{J}_k} \\ &\leq \sum_{k=0}^{k=N} \min_{i=1}^{i=N} L_{\mathcal{J}_{k,i}} + \sum_{k=0}^{k=N} \sqrt{\frac{2^k \log N}{2}} \\ &\leq \min_{i=1}^{i=N} L_{T,i} + \sqrt{\frac{\log N}{2}} \sum_{k=0}^{k=N} 2^{\frac{k}{2}} \end{aligned}$$

Proof of Theorem 8.7



- ▶ The second inequality on the preceding slide uses the property of super-additivity of min which means $\min_i X_i + \min_i Y_i \leq \min_i X_i + Y_i$.

- ▶ This implies that
$$\sum_{k=0}^{k=n} \min_{i=1}^{i=N} L_{\mathcal{J}_k, i} \leq \min_{i=1}^{i=N} \sum_{k=0}^{k=n} L_{\mathcal{J}_k, i}.$$

- ▶ The geometric sum appearing on the right hand side of the last inequality on the previous slide can be simplified to

$$\sum_{k=0}^{k=N} 2^{\frac{k}{2}} = \frac{2^{\frac{n+1}{2}} - 1}{\sqrt{2} - 1} \leq \frac{\sqrt{2}(\sqrt{T+1}) - 1}{\sqrt{2} - 1} \leq \frac{\sqrt{2}(\sqrt{T+1}) - 1}{\sqrt{2} - 1} = \frac{\sqrt{2}\sqrt{T}}{\sqrt{2} - 1} + 1.$$

- ▶ Plugging this back into the last inequality on the previous slide and simplifying leads to the expression in the statement of the theorem.

Rademacher complexity



- ▶ The hypothesis sets used in machine learning are generally infinite.
- ▶ The sample complexity bounds proposed in the lectures so far deal with finite hypothesis sets.
- ▶ We do have examples (axis-aligned rectangles) of hypothesis sets that are PAC-learnable even though the number of possible hypotheses is infinite.
- ▶ We would like to devise learning guarantees for infinite hypothesis sets.

Rademacher complexity



- ▶ A general idea for dealing with infinite hypothesis sets is to reduce such problems to finite hypothesis sets and use the techniques developed so far.
- ▶ There are different techniques for such reductions, each relying on different notions of complexity for such hypothesis sets.
- ▶ We will explore one such complexity notion called Rademacher complexity in this lecture.
- ▶ This will help us derive learning guarantees using proofs based on McDiarmid's inequality.

Rademacher complexity



- ▶ Let $\mathcal{H}$ denote a hypothesis set.
- ▶ Let $L : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$ be an arbitrary loss function.
- ▶ Let $\mathcal{G}$ denote a family of loss functions associated to $\mathcal{H}$ mapping from $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$ to $\mathbb{R}$. Thus $\mathcal{G} = \{g : (x, y) \rightarrow L(h(x), y) : h \in \mathcal{H}\}$.
- ▶ Informally the Rademacher complexity captures the richness of a hypothesis set by measuring how well the hypothesis set fits random noise. The formal definition is given in the next slide.

Definition 3.1 - Empirical Rademacher complexity



Let $\mathcal{G}$ be a family of functions mapping from $\mathcal{Z}$ to $[a, b]$ and $S = (z_1, z_2, \dots, z_m)$ a fixed sample of size m with elements in $\mathcal{Z}$. The empirical Rademacher complexity of $\mathcal{G}$ with respect to the sample S is defined as $\hat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}_{\sigma} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m \sigma_i g(z_i) \right]$ where $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_m)^T$ with σ_i s as independent uniform random variables taking their values in $\{-1, 1\}$. the random variables σ_i are called Rademacher variables.

Empirical Rademacher complexity



- ▶ Let $\mathbf{g}_S$ denote the vector of values taken by the function g over the sample S , i.e $\mathbf{g}_S = (g(z_1), g(z_2), \dots, g(z_m))^T$.
- ▶ The empirical Rademacher complexity can be rewritten as
$$\hat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}_{\sigma} \left[\sup_{g \in \mathcal{G}} \frac{\sigma \cdot \mathbf{g}_S}{m} \right].$$
- ▶ The inner product $\frac{\sigma \cdot \mathbf{g}_S}{m}$ measures the correlation of $\mathbf{g}_S$ with the random noise σ .
- ▶ The supremum $\sup_{g \in \mathcal{G}} \frac{\sigma \cdot \mathbf{g}_S}{m}$ is a measure of how well the function class $\mathcal{G}$ correlates with σ over the sample S .

Definition 3.2 - Rademacher complexity



Let $\mathcal{D}$ denote the distribution according to which samples are drawn. For any integer $m \geq 1$ the Rademacher complexity of $\mathcal{G}$ is the expectation of the empirical Rademacher complexity over all samples of size m drawn according to $\mathcal{D}$. We have

$$\mathfrak{R}_m(\mathcal{G}) = \mathbb{E}_{S \sim \mathcal{D}} [\hat{\mathfrak{R}}_S(\mathcal{G})]$$

Theorem 3.3



Let $\mathcal{G}$ be a family of functions mapping from $\mathcal{Z}$ to $[0, 1]$. Then for any $\delta > 0$, with probability at least $1 - \delta$ over the draw of any i.i.d sample S of size m , each of the following holds for all $g \in \mathcal{G}$.

$$\mathbb{E}[g(z)] \leq \frac{1}{m} \sum_{i=1}^m g(z_i) + 2\mathfrak{R}_m(\mathcal{G}) + \sqrt{\frac{\log \frac{1}{\delta}}{2m}}$$

$$\mathbb{E}[g(z)] \leq \frac{1}{m} \sum_{i=1}^m g(z_i) + 2\hat{\mathfrak{R}}_S(\mathcal{G}) + 3\sqrt{\frac{\log \frac{1}{\delta}}{2m}}$$



- ▶ The proof of Theorem 3.3 is fairly involved and we need some mathematical preliminaries before we embark on it.
- ▶ In the following slides we shall provide these preliminaries.
- ▶ We need in particular a result on the behaviour of the supremum of a sequence of values.
- ▶ We shall need the concepts of sub-additivity of the supremum function and Rademacher variables.

Preliminaries for Theorem 3.3



We shall first demonstrate the sub-additivity of the supremum function. This means that the inequality below is true:

$$\sup(A_1 + B_1, A_2 + B_2) \leq \sup(A_1, A_2) + \sup(B_1, B_2)$$

Proof:

$$\begin{aligned} \sup(A_1 + B_1, A_2 + B_2) &\leq \sup(A_1 + B_1, A_1 + B_2, A_2 + B_2, A_2 + B_1) \\ &\leq \sup(\sup(A_1 + B_1, A_1 + B_2), \sup(A_2 + B_1, A_2 + B_2)) \\ &\leq \sup(A_1 + \sup(B_1, B_2), A_2 + \sup(B_1, B_2)) \\ &\leq \sup(A_1, A_2) + \sup(B_1, B_2) \end{aligned}$$

We can extend the above to

$$\sup(A_1 + B_1, \dots, A_m + B_m) \leq \sup(A_1, \dots, A_m) + \sup(B_1, \dots, B_m).$$

Discrete version of Jensen's inequality



Let $\lambda \in (0, 1)$. We shall show that

$$\sup(\lambda A + (1 - \lambda)B, \lambda C + (1 - \lambda)D) \leq \lambda \sup(A, C) + (1 - \lambda) \sup(B, D).$$

$$\begin{aligned} & \sup(\lambda A + (1 - \lambda)B, \lambda C + (1 - \lambda)D) \\ & \leq \sup(\lambda A + (1 - \lambda)B, \lambda A + (1 - \lambda)D, \lambda C + (1 - \lambda)D, \lambda C + (1 - \lambda)B) \\ & \leq \sup(\sup(\lambda A + (1 - \lambda)B, \lambda A + (1 - \lambda)D), \sup(\lambda C + (1 - \lambda)D, \lambda C + (1 - \lambda)B)) \\ & \leq \sup(\lambda A + (1 - \lambda) \sup(B, D), \lambda C + (1 - \lambda) \sup(B, D)) \\ & \leq \lambda \sup(A, C) + (1 - \lambda) \sup(B, D) \end{aligned}$$

Discrete version of Jensen's inequality



- ▶ We can interpret A and B in the preceding slide as values taken by a random variable $f_1(X)$, where X is a discrete random variable that takes exactly two values with probabilities λ and $1 - \lambda$ respectively. Thus $\lambda A + (1 - \lambda)B$ can be viewed as the expectation of $f_1(X)$, $\mathbb{E}_X(f_1(X))$.
- ▶ Similarly we can interpret C and D in the preceding slide as values taken by a random variable $f_2(X)$ where X is the same random variable sampled in the previous bullet point. Thus $\lambda C + (1 - \lambda)D$ can be viewed as the expectation of $f_2(X)$, $\mathbb{E}_X(f_2(X))$.

Discrete version of Jensen's inequality



- ▶ Thus the inequality proved in the preceding slide amounts to the following:
$$\sup(\mathbb{E}_X(f_1(X)), \mathbb{E}_X(f_2(X))) \leq \mathbb{E}_X(\sup(f_1(X), f_2(X))).$$
- ▶ The inequality can be informally stated as the suprema of expectations is less than or equal to the expectation of suprema.
- ▶ It is a version of Jensen's inequality which states for convex functions f , $f(E(X)) \leq E(f(X))$. "sup" is a convex function.

Proof of Theorem 3.3



- ▶ For any sample $S = (z_1, z_2, \dots, z_m)$ and any $g \in \mathcal{G}$ let $\hat{E}_S[g]$ denote the empirical average of g over S , ie $\hat{E}_S[g] = \frac{1}{m} \sum_{i=1}^m g(z_i)$.
- ▶ The proof consists of applying McDiarmid's inequality to the quantity $\phi(S) = \sup_{g \in \mathcal{G}} (\mathbb{E}[g] - \hat{E}_S[g])$.
- ▶ Let S and S' be two samples of size m that differ in exactly one point, say z_m in S becomes z'_m in S' .

Proof of Theorem 3.3



- ▶ From the sub-additivity of the supremum function shown a few slides ago we see that $\sup(A - C, B - D) + \sup(C, D) \geq \sup(A, B)$, or that $\sup(A - C, B - D) \geq \sup(A, B) - \sup(C, D)$. This means that the difference of suprema does not exceed the supremum of differences.
- ▶ Applying this to $\phi(S)$ and $\phi(S')$ we have

$$\begin{aligned}\phi(S') - \phi(S) &\leq \sup_{g \in \mathcal{G}} (\hat{\mathbb{E}}_S[g] - \hat{\mathbb{E}}_{S'}[g]) \\ &= \sup_{g \in \mathcal{G}} \frac{g(z_m) - g(z'_m)}{m} \leq \frac{1}{m}\end{aligned}$$

Proof of Theorem 3.3



- ▶ Similarly we can obtain $\phi(S) - \phi(S') \leq \frac{1}{m}$, and thus $|\phi(S) - \phi(S')| \leq \frac{1}{m}$.
- ▶ Then by McDiarmid's inequality, with probability at least $1 - \frac{\delta}{2}$, the following holds:

$$\phi(S) \leq \mathbb{E}[\phi(S)] + \sqrt{\frac{\log \frac{2}{\delta}}{2m}}.$$

- ▶ We shall now use a series of non-trivial transformations to bound the quantity on the right hand side of the above inequality in succeeding slides.

Proof of Theorem 3.3



$$\mathbb{E}_S[\phi(S)] = \mathbb{E}_S \left[\sup_{g \in \mathcal{G}} (\mathbb{E}[g] - \hat{\mathbb{E}}_S[g]) \right]$$

$$\mathbb{E}_S[\phi(S)] = \mathbb{E}_S \left[\sup_{g \in \mathcal{G}} (\mathbb{E}_{S'}[\hat{\mathbb{E}}_{S'}[g] - \hat{\mathbb{E}}_S[g]) \right]$$

Here we are using the fact that $\mathbb{E}[g] = \mathbb{E}_{S'}[\hat{\mathbb{E}}_{S'}[g]]$.

Proof of Theorem 3.3



$$\mathbb{E}_S \left[\sup_{g \in \mathcal{G}} \mathbb{E}_{S'} (\hat{\mathbb{E}}_{S'}[g] - \hat{\mathbb{E}}_S[g]) \right] \leq \mathbb{E}_{S, S'} \left[\sup_{g \in \mathcal{G}} (\hat{\mathbb{E}}_{S'}[g] - \hat{\mathbb{E}}_S[g]) \right]$$

Here we are using the fact that the supremum of expectations is less than or equal to the expectation of suprema. This was proved as a part of the mathematical preliminaries needed for this theorem.

Proof of Theorem 3.3



$$\begin{aligned}\mathbb{E}_{S,S'} \left[\sup_{g \in \mathcal{G}} (\hat{\mathbb{E}}_{S'}[g] - \hat{\mathbb{E}}_S[g]) \right] &= \mathbb{E}_{S,S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{i=m} (g(z'_i) - g(z_i)) \right] \\ &= \mathbb{E}_{\sigma,S,S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{i=m} \sigma_i (g(z'_i) - g(z_i)) \right]\end{aligned}$$

Proof of Theorem 3.3



- ▶ Here we use Rademacher variables σ_i which are uniform random variables which take values from the set $\{-1, 1\}$. The use of Rademacher variables does not change the expectation.
- ▶ Let us examine closely why the use of Rademacher variables does not change the expectation.
- ▶ Let S and S' be two samples of size m respectively. Let us form a set K of tuples as follows:
 $\{(S_1, S'_1), (S_2, S'_2), \dots, (S_m, S'_m)\}$.
- ▶ Let $\mathbf{u}$ denote a string of length m where the i th entry is 1 or -1, and $\mathbf{u}'$ denote the complementary string, ie. if $\mathbf{u}$ has a 1 in the i th position, then $\mathbf{u}'$ has a -1 in the i th position and vice-versa.

Proof of Theorem 3.3



- ▶ Let K_u denote a set that is composed out of K as follows: if $u_i = 1$, K_u picks the first entry in the i th tuple of K . Otherwise it picks the second entry in the i th tuple.
- ▶ Note that the original expectation expression has terms of the form $P(S)P(S')f(S, S')$.
- ▶ For each set K_u and $K_{u'}$, there is exactly one way to swap elements between K_u and $K_{u'}$ so as to turn K_u into the given set S . Let us denote the swapping operation in terms of a string $\sigma(K_u)$ whose i th entry is 1 if we do not need to swap out the i th element of K_u and -1 if we need to swap out the i th element of K_u .

Proof of Theorem 3.3



- ▶ The second expectation expression has terms such as $P(\sigma)P(K_u))P(K_{u'})f(S, S')$.
- ▶ We see that $P(K_u) = P(S) = P(S') = P(K_{u'})$ since all are samples of length m that have been sampled i.i.d, and $P(\sigma) = \frac{1}{2^m}$.
- ▶ There are 2^m strings of the form K_u which can each be transformed into S using appropriately defined Rademacher variables leading to 2^m terms which all sum to $P(S)P(S')f(S, S')$. Thus the two expectations are equal.

Proof of Theorem 3.3



$$\begin{aligned} \mathbb{E}_{\sigma, S, S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{i=m} \sigma_i(g(z'_i) - g(z_i)) \right] &\leq \mathbb{E}_{\sigma, S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{i=m} \sigma_i(g(z'_i)) \right] \\ -\mathbb{E}_{\sigma, S} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{i=m} \sigma_i(g(z_i)) \right] &= 2\mathbb{E}_{\sigma, S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{i=m} \sigma_i(g(z'_i)) \right] = 2\mathfrak{R}_m(\mathcal{G}). \end{aligned}$$

Here we use the sub-additivity of the supremum function and the fact that the variables σ_i and $-\sigma_i$ are distributed in the same way.

$$\mathbb{E}[g(z)] - \frac{1}{m} \sum_{i=1}^m g(z_i) = \mathbb{E}[g(z)] - \hat{\mathbb{E}}[g] \leq \sup_{g \in \mathcal{G}} (\mathbb{E}[g] - \hat{\mathbb{E}}_S[g]) = \phi(S)$$

$$\phi(S) \leq \mathbb{E}_S[\phi(S)] + \sqrt{\frac{\log \frac{2}{\delta}}{2m}}$$

$$\mathbb{E}_S[\phi(S)] = 2\mathfrak{R}_m(\mathcal{G})$$

$$\mathbb{E}[g(z)] = \frac{1}{m} \sum_{i=1}^m g(z_i) + 2\mathfrak{R}_m(\mathcal{G}) + \sqrt{\frac{\log \frac{2}{\delta}}{2m}}$$

Proof of Theorem 3.3



- ▶ To derive a bound in terms of $\hat{\mathfrak{R}}_S(\mathcal{G})$ we note that when S and S' differ by one point, the difference between $\hat{\mathfrak{R}}_S(\mathcal{G})$ and $\hat{\mathfrak{R}}_{S'}(\mathcal{G})$ is less than $1/m$.

- ▶ By McDiarmid's inequality this leads to

$$\mathfrak{R}_m(\mathcal{G}) \leq \hat{\mathfrak{R}}_S(\mathcal{G}) + \sqrt{\frac{\log \frac{2}{\delta}}{2m}}.$$

- ▶ Finally by the union bound we obtain

$$\phi(S) \leq 2\hat{\mathfrak{R}}_S(\mathcal{G}) + 3\sqrt{\frac{\log \frac{2}{\delta}}{2m}}.$$

Lemma 3.4



Let $\mathcal{H}$ be a family of functions taking values in $\{-1, +1\}$ and let $\mathcal{G}$ be the family of loss functions associated to $\mathcal{H}$ for the zero-one loss: $\mathcal{G} = \{(x, y) \rightarrow 1_{h(x) \neq y} : h \in \mathcal{H}\}$. For any sample $S = ((x_1, y_1), (x_2, y_2), \dots, (x_m, y_m))$ of elements in $\mathcal{X} \times \{+1, -1\}$, let $S_{\mathcal{X}}$ denote its projection over $\mathcal{X}$: $S_{\mathcal{X}} = (x_1, \dots, x_m)$. Then the following relation holds between the empirical Rademacher complexities of $\mathcal{G}$ and $\mathcal{H}$:

$$\hat{\mathfrak{R}}_S(\mathcal{G}) = \frac{1}{2} \hat{\mathfrak{R}}_{S_{\mathcal{X}}}(\mathcal{H})$$

Proof of Lemma 3.4



For any sample $S = ((x_1, y_1), (x_2, y_2), \dots, (x_m, y_m))$ of elements in $\mathcal{X} \times \{+1, -1\}$, we can write the empirical Rademacher complexity as follows:

$$\begin{aligned}\hat{\mathfrak{R}}_S(\mathcal{G}) &= \mathbb{E}_{\sigma} \left[\sup_{h \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m \sigma_i 1_{h(x_i) \neq y_i} \right] = \mathbb{E}_{\sigma} \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i \frac{1 - y_i h(x_i)}{2} \right] \\ &= \frac{1}{2} \mathbb{E}_{\sigma} \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m -\sigma_i y_i h(x_i) \right] \\ &= \frac{1}{2} \mathbb{E}_{\sigma} \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i h(x_i) \right] = \frac{1}{2} \hat{\mathfrak{R}}_{S_{\mathcal{X}}}(\mathcal{H})\end{aligned}$$

Theorem 3.5 - Rademacher complexity bounds



Let $\mathcal{H}$ be a family of functions taking values in $\{-1, +1\}$ and let D be the distribution over the input space $\mathcal{X}$. Then for any $\delta > 0$, with probability at least $1 - \delta$ over a sample S of size m according to D , each of the following holds for any $h \in \mathcal{H}$.

$$R(h) \leq \hat{R}_S(h) + \mathfrak{R}_m(\mathcal{H}) + \sqrt{\frac{\log \frac{1}{\delta}}{2m}}$$

$$R(h) \leq \hat{R}_S(h) + \mathfrak{R}_S(\mathcal{H}) + 3\sqrt{\frac{\log \frac{2}{\delta}}{2m}}$$

Proof of Theorem 3.5



- ▶ Note that for binary classification $\mathbb{E}[g(z)] = R(h)$.
- ▶ Theorem 3.3 gives us results in terms of $\mathbb{E}[g(z)]$.
- ▶ Lemma 3.4 allows us to replace $\mathfrak{R}_m(\mathcal{G})$ with $\frac{1}{2}\mathfrak{R}_m(\mathcal{H})$ and $\hat{\mathfrak{R}}_S(\mathcal{G})$ with $\frac{1}{2}\mathfrak{R}_S(\mathcal{H})$ to get the statement of Theorem 3.5.

Using Theorem 3.5



- ▶ The second inequality of Theorem 3.5 is data-dependent in that we have an upper bound on the generalization error in terms of quantities that can be calculated from a particular sample.
- ▶ If we could compute $\hat{\mathcal{R}}_S(\mathcal{H})$, this bound could be particularly informative.
- ▶ What is the complexity involved in computing $\hat{\mathcal{R}}_S(\mathcal{H})$?

Using Theorem 3.5



- Using the fact that $-\sigma_i$ and σ_i are distributed in the same way, we can write

$$\begin{aligned}\hat{\mathcal{R}}_S(\mathcal{H}) &= \frac{1}{2\sigma} \mathbb{E} \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i y_i h(x_i) \right] \\ &= \frac{1}{2\sigma} \mathbb{E} \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m -\sigma_i y_i h(x_i) \right] \\ &= -\frac{1}{2\sigma} \mathbb{E} \left[\inf_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i y_i h(x_i) \right]\end{aligned}$$

Using Theorem 3.5



- ▶ For a fixed value of σ computing the last quantity in the previous slide is an empirical risk minimization problem which is known to be computationally hard for some hypothesis sets.
- ▶ Thus computing $\hat{\mathcal{R}}_S(\mathcal{H})$ could be hard for some hypothesis sets.
- ▶ We will now relate the Rademacher complexity to some combinatorial measures which are easier to compute and also of independent interest in the analysis of learning in different contexts.

Definition 3.6 - Growth function



The growth function $\Pi_{\mathcal{H}} : \mathbb{N} \rightarrow \mathbb{N}$ for a hypothesis set $\mathcal{H}$ is defined by

$$\forall m \in \mathbb{N}, \Pi_{\mathcal{H}}(m) = \max_{\{x_1, x_2, \dots, x_m\} \subset \mathcal{X}} |\{(h(x_1), h(x_2), \dots, h(x_m)) : h \in \mathcal{H}\}|.$$

In other words $\Pi_{\mathcal{H}}(m)$ is the maximum number of distinct ways in which m points can be classified using hypotheses in $\mathcal{H}$.

Growth function



- ▶ Each one of the distinct classifications of the m points represents a dichotomy.
- ▶ The growth function counts the number of dichotomies that are realized by the hypothesis.
- ▶ This provides another measure of the richness of the hypothesis set $\mathcal{H}$.
- ▶ Unlike the Rademacher complexity, it does not depend on the distribution. It is purely combinatorial.

Theorem 8.7 - Massart's lemma



Let $A \subset \mathbb{R}^m$ be a finite set with $r = \max_{\mathbf{x} \in A} \|\mathbf{x}\|_2$, then the following holds:

$$\mathbb{E}[\sup_{\mathbf{x} \in A} \frac{1}{m} \sum_{i=1}^m \sigma_i x_i] \leq \frac{r \sqrt{2 \log |A|}}{m}$$

Proof of Theorem 8.7



- ▶ The proof of the theorem follows from a bound on the expectation of a maximum.
- ▶ The random variables $\sigma_i x_i$ are independent.
- ▶ Each $\sigma_i x_i$ takes values in $-|x_i|, |x_i|$.
- ▶ We can then apply Corollary D.11 from Foundations of Machine Learning, 2nd Edition to complete the argument.

Corollary 3.8



Let $\mathcal{G}$ be a family of functions taking values in $\{-1, +1\}$. Then the following holds:

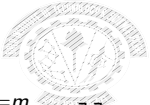
$$\mathfrak{R}_m(\mathcal{G}) \leq \sqrt{\frac{2 \log \Pi_{\mathcal{G}}(m)}{m}}$$

Thus the result above bounds the Rademacher complexity in terms of the growth function.

Proof of Corollary 3.8



For a fixed sample $S = (x_1, x_2, \dots, x_m)$ let us denote by $\mathcal{G}_{|S}$ the set of vectors of function values $(g(x_1), g(x_2), \dots, g(x_m))^T$ where g is in $\mathcal{G}$. Since $g \in \mathcal{G}$ takes values in $\{-1, +1\}$, the norm of these vectors is bounded by $\sqrt{m}$. We can then apply Massart's lemma as follows:


$$\mathfrak{R}_m(\mathcal{G}) = \mathbb{E}_S \left[\mathbb{E}_\sigma \left[\frac{1}{m} \sum_{\substack{i=1 \\ u \in \mathcal{G}_{|S}}}^m \sigma_i u_i \right] \right] \leq \mathbb{E}_S \left[\frac{\sqrt{m} \sqrt{2 \log |\mathcal{G}_{|S}|}}{m} \right]$$

Proof of Corollary 3.8



By definition $G|_S$ is bounded by the growth function. Thus we can write the equations in the previous slide as follows:

$$\mathfrak{R}_m(\mathcal{G}) \leq \mathbb{E}_S \left[\frac{\sqrt{m} \sqrt{2 \log |G|_S}}{m} \right]$$
$$\mathbb{E}_S \left[\frac{\sqrt{m} \sqrt{2 \log |G|_S}}{m} \right] \leq \mathbb{E}_S \left[\frac{\sqrt{m} \sqrt{2 \log \Pi_{\mathcal{G}}(m)}}{m} \right] = \sqrt{\frac{2 \log \Pi_{\mathcal{G}}(m)}{m}}$$

Corollary 3.9



Combining the generalization bound from Theorem 3.5 with corollary 3.8 yields Corollary 3.9.

Let $\mathcal{H}$ be a family of functions taking values in $\{-1, +1\}$. Then for any $\delta > 0$ with probability at least $1 - \delta$, for any $h \in \mathcal{H}$,

$$R(h) \leq \hat{R}_S(h) + \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}} + \sqrt{\frac{\log \frac{1}{\delta}}{2m}}$$

Vapnik-Chervonenkis (VC) dimension



- ▶ We will look at the VC-dimension in this lecture.
- ▶ Like Rademacher complexity it is also a combinatorial measure, but it is often easier to compute than it.
- ▶ We have already seen the concept of shattering in order to derive the lower bound for the mistake model in online learning.
- ▶ Let us have a look at it again.



- ▶ For a given hypothesis set $\mathcal{H}$, a dichotomy of a set S is one of the ways of labelling the elements in S using a hypothesis in $\mathcal{H}$.
- ▶ A set S of $m > 1$ points is said to be shattered by a hypothesis set $\mathcal{H}$, when $\mathcal{H}$ realizes all possible dichotomies of S . In other words for any assignment of labels to the points in S , there exists a hypothesis $h \in \mathcal{H}$ that achieves that labelling.
- ▶ When a set S of size $m > 1$ is shattered by a hypothesis set $\mathcal{H}$, we see that $\Pi_{\mathcal{H}}(m) = 2^m$.
- ▶ The VC-dimension of a hypothesis set $\mathcal{H}$ is the size of the largest set that can be shattered by $\mathcal{H}$.

$$VCdim(\mathcal{H}) = \max\{m : \Pi_{\mathcal{H}}(m) = 2^m\}$$

Unrealizable dichotomies

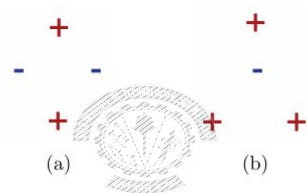


Figure 3.2

Unrealizable dichotomies for four points using hyperplanes in $\mathbb{R}^2$. (a) All four points lie on the convex hull. (b) Three points lie on the convex hull while the remaining point is interior.

Unrealizable dichotomies



- ▶ Note that in the figure on the previous slide, any hyperplane will classify all points on one side as positive and points on the other side as negative.
- ▶ No matter what alignment is chosen for a given hyperplane, it will end up classifying some of the points wrongly.
- ▶ Therefore the two dichotomies shown are unrealizable for any hypothesis set consisting only of hyperplanes.
- ▶ Note that when we declare that the VC-dimension of a hypothesis set is d , it does not mean that all sets of d points are shattered. It suffices to have only one set of d points that can be shattered.

Computing the VC-dimension



- ▶ To compute the VC-dimension of a hypothesis set $\mathcal{H}$ we will first show a lower bound for its value and then find an upper bound.
- ▶ To give a lower bound on $VCdim(\mathcal{H})$ it suffices to exhibit a set of size d that can be shattered by $\mathcal{H}$.
- ▶ To show an upper bound, we need to show that no set of size $d + 1$ can be shattered by $\mathcal{H}$. This task is more difficult.
- ▶ Next we will study intervals on a real line and calculate the VC-dimension of this hypothesis set.

VC-dimension of intervals on the real line

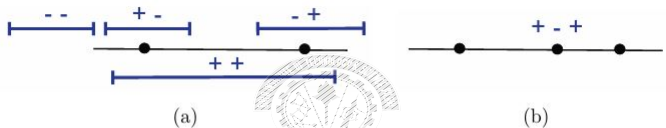


Figure 3.1

VC-dimension of intervals on the real line. (a) Any two points can be shattered. (b) No sample of three points can be shattered as the $(+, -, +)$ labeling cannot be realized.

VC-dimension of intervals on the real line



- ▶ In the first figure on the left hand side we see that all four dichotomies can be realized for a set of two points i.e. $(+, +), (+, -), (-, +), (-, -)$ where we respectively come up with an interval that encloses both given points, excludes the second one only, excludes the first one only or excludes both points.
- ▶ However it is not possible for an interval on the real line to shatter a set of 3 points - in particular the dichotomy $(+, -, +)$ cannot be realized.
- ▶ This is because an interval is a convex region - including points A and B in the interval forces the inclusion of all points in between.
- ▶ Thus in this case it is easy to see that the VC-dimension of the hypothesis set consisting of intervals on a real line is 2.

VC-dimension of hyperplanes



- ▶ Any three non-collinear points in $\mathbb{R}^2$ can be shattered.
- ▶ To see this assume that all three points are labelled positive - we can draw a hyperplane so that all the three points are on one side of the hyperplane and label that side positive. Similarly for the case when all three points are negative.
- ▶ When some two of the points are of one sign, and the other point is of the opposite sign we can draw a hyperplane so as to achieve this dichotomy as well.

VC-dimension of hyperplanes



- ▶ We have already seen in a previous slide that no set of 4 points can be shattered.
- ▶ We did this by considering that the set of 4 points will define a convex hull or the 4th point would lie inside the convex hull of the first three. In each case there is a particular labelling that cannot be realised by any hyperplane hypothesis.
- ▶ Thus $VCdim(\mathcal{H}) = 3$ when $\mathcal{H} =$ hyperplanes in $\mathbb{R}^2$.
- ▶ Let us examine the case of hyperplanes in $\mathbb{R}^d$.

VC-dimension of hyperplanes in $\mathbb{R}^d$



- ▶ We shall show that there exists a set of $d + 1$ points which can be shattered by the hypothesis set consisting of hyperplanes in $\mathbb{R}^d$. This will establish a lower bound for the VC-dimension of this hypothesis set.
- ▶ Let us set $\mathbf{x}_0$ to be the origin and each $\mathbf{x}_i, 1 \leq i \leq d$ to be the i th canonical vector, ie a 1 in the i th position and zeroes everywhere else.
- ▶ Let $y_0, y_1, \dots, y_{d+1} \in \{-1, +1\}$ be an arbitrary set of labels for $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_d$.
- ▶ Let $\mathbf{w}$ be a classifier whose i th coordinate is y_i .

VC-dimension of hyperplanes in $\mathbb{R}^d$



The classifier defined by the equation $\mathbf{w} \cdot \mathbf{x} + \frac{y_0}{2} = 0$ shatters $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_d$ since for any $i \in \{0, 1, \dots, d\}$

$$\text{sgn}(\mathbf{w} \cdot \mathbf{x}_i + \frac{y_0}{2}) = \text{sgn}(y_i + \frac{y_0}{2}) = y_i$$

To obtain an upper bound we will show next that no set of $d + 2$ points can be shattered by halfspaces in $\mathbb{R}^d$.



Theorem 3.13 Any set $\mathcal{X}$ of $d + 2$ points in $\mathbb{R}^d$ can be partitioned into two subsets $\mathcal{X}_1$ and $\mathcal{X}_2$ such that the convex hulls of $\mathcal{X}_1$ and $\mathcal{X}_2$ intersect.

Proof of Radon's theorem



- ▶ Let $\mathcal{X} = \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{d+2} \subset \mathbb{R}^d$.
- ▶ First we set up the following system of $d + 1$ linear equations in $d + 2$ variables $\alpha_1, \alpha_2, \dots, \alpha_{d+2}$.

$$\sum_{i=1}^{i=d+2} \alpha_i \mathbf{x}_i = 0$$

$$\sum_{i=1}^{i=d+2} \alpha_i = 0$$

Proof of Radon's theorem



- ▶ The system of $d + 1$ equations in $d + 2$ variables on the previous slide can be written in matrix form as $Ax = b$ where A is a $(d + 1) \times (d + 2)$ matrix.
- ▶ We then see that A has rank no greater than $d + 1$ which means there is at least one non-zero solution in its nullspace.
- ▶ Let us call this solution $\beta_1, \beta_2, \dots, \beta_{d+2}$.
- ▶ Since $\sum_{i=1}^{d+2} \beta_i = 0$ from the last equation on the previous slide, we can see that $\mathcal{J}_1 = \{i \in [d + 2] : \beta_i > 0\}$ and $\mathcal{J}_2 = \{i \in [d + 2] : \beta_i \leq 0\}$ are non-empty sets.

Proof of Radon's theorem



► Let $\mathcal{X}_1 = \{\mathbf{x}_i, i \in \mathcal{J}_1\}$ and $\mathcal{X}_2 = \{\mathbf{x}_i, i \in \mathcal{J}_2\}$ for a partition of $\mathcal{X}$.

► Also we have $\sum_{i \in \mathcal{J}_1} \beta_i = -\sum_{i \in \mathcal{J}_2} \beta_i$.

► Let $\beta = \sum_{i \in \mathcal{J}_1} \beta_i$.

► From the equation $\sum_{i=1}^{i=d+2} \alpha_i \mathbf{x}_i = 0$, we can write $\sum_{i \in \mathcal{J}_1} \frac{\beta_i}{\beta} \mathbf{x}_i = -\sum_{i \in \mathcal{J}_2} \frac{\beta_i}{\beta} \mathbf{x}_i$ with $\sum_{i \in \mathcal{J}_1} \frac{\beta_i}{\beta} = \sum_{i \in \mathcal{J}_1} \frac{\beta_i}{\beta} = 1$.

► We also have each $\frac{\beta_i}{\beta} > 0$ for $i \in \mathcal{J}_1$ and $-\frac{\beta_i}{\beta} > 0$ for $i \in \mathcal{J}_2$.

Thus we show that the convex combination $\sum_{i \in \mathcal{J}_1} \frac{\beta_i}{\beta} \mathbf{x}_i$ belongs both to the convex hull of $\mathcal{X}_1$ and the convex hull of $\mathcal{X}_2$.

VC-dimension of hyperplanes in $\mathbb{R}^d$



- ▶ We now need to complete the argument relating to the VC-dimension of the hypothesis set consisting of hyperplanes in $\mathbb{R}^d$.
- ▶ By Radon's theorem, if we are given a set $\mathcal{X}$ of $d + 2$ points, it can be partitioned into two sets $\mathcal{X}_1$ and $\mathcal{X}_2$ so that their convex hulls intersect.
- ▶ When two sets $\mathcal{X}_1$ and $\mathcal{X}_2$ are separated by a hyperplane, their convex hulls are also separated by the hyperplane.
- ▶ When two convex hulls intersect they cannot be separated by any hyperplane. Thus the sets $\mathcal{X}_1$ and $\mathcal{X}_2$ are not separated by any hyperplane and $\mathcal{X}$ is not shattered.
- ▶ Combining the lower and upper bounds we have shown that $VCdim(\text{hyperplanes in } \mathbb{R}^d) = d + 1$.

Theorem 3.17 (Sauer's lemma)



Let $\mathcal{H}$ be a hypothesis set with $VCdim(\mathcal{H}) = d$. Then, for all $m \in \mathbb{N}$, the following holds:

$$|\Pi_{\mathcal{H}}(m)| \leq \sum_{i=0}^{i=d} \binom{m}{i}$$

Proof of Sauer's lemma



- ▶ We shall prove the theorem using induction on $m + d$. The theorem is clearly true when $m = 0$ and $d = 0$ or $d = 1$.
- ▶ Assume that it holds for $(m - 1, d - 1)$ and $(m - 1, d)$.
- ▶ Fix a set $S = \{x_1, x_2, \dots, x_m\}$ with $\Pi_H(m)$ dichotomies and let $\mathcal{G} = \mathcal{H}|_S$ be the set of concepts $\mathcal{H}$ induced by restriction to S .

Proof of Sauer's lemma



- ▶ Consider the set $S' = \{x_1, x_2, \dots, x_{m-1}\}$. Define $\mathcal{G}_1 = \mathcal{G}|_{S'}$ as the set of concepts $\mathcal{H}$ induced by the restriction to S' .
- ▶ We identify each concept as the set of samples in S or S' for which the concept returns a non-zero value. With this definition we can define $\mathcal{G}_2 = \{g' \subset S' : (g' \in \mathcal{G}) \wedge (g' \cup x_m \in \mathcal{G})\}$.
- ▶ Let us analyse what kind of concepts belong to $\mathcal{G}_2$. Since $g' \subset S'$, $g' \in \mathcal{G}$ means it is a concept of $\mathcal{G}$ without adding x_m . However when we add x_m to g' it remains a concept of $\mathcal{G}$.
- ▶ The construction of $\mathcal{G}_1$ and $\mathcal{G}_2$ is shown in the figure on the next slide.

Proof of Sauer's lemma



$$\mathcal{G}_1 = \mathcal{G}|_{S'} \quad \mathcal{G}_2 = \{g' \subseteq S' : (g' \in \mathcal{G}) \wedge (g' \cup \{x_m\} \in \mathcal{G})\}$$

x_1	x_2	$\dots$	x_{m-1}	x_m
1	1	0	1	0
1	1	0	1	1
0	1	1	1	1
1	0	0	1	0
1	0	0	0	1
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$

Figure 3.6

Illustration of how $\mathcal{G}_1$ and $\mathcal{G}_2$ are constructed in the proof of Sauer's lemma.

Proof of Sauer's lemma



- ▶ In the figure on the previous slide each row represents the behaviour of a hypothesis on the given samples $x_1, x_2, \dots, x_m$, i.e the 1s and 0s seen in the i th row are $h_i(x_1), h_i(x_2), \dots, h_i(x_m)$.
- ▶ Each concept is represented in terms of the set of samples for which the concept output is a 1. For example, the first row of the table in the figure on the previous slide shows $h_1(x_1) = 1, h_1(x_2) = 1, h_1(x_4) = 1$.
- ▶ The concept h_1 satisfies $h_1 \in S'$ which means that all the samples at which h_1 returns 1 belong to S' , and $h_1 \in \mathcal{G}$, i.e it is a concept of $\mathcal{G}$ even without having a 1 for the sample x_m . If we add x_m to h_1 it remains a concept of $\mathcal{G}$. Thus $h_1 \in \mathcal{G}_1$. Note that this property does not hold for the other concepts that are shown to belong exclusively to $\mathcal{G}_1$.

Proof of Sauer's lemma



- ▶ Consider a concept c in $\mathcal{G}$ - assume it is of the form $p(c)0$ where $p(c)$ is a string of ones and zeroes representing the behaviour of the hypothesis h on the samples in S' and $h(x_m) = 0$. If $\mathcal{G}$ also contains a concept c' of the form $p(c)1$ where $h(x_m) = 1$, then the representation $p(c)$ belongs to both $\mathcal{G}_1$ and $\mathcal{G}_2$.
- ▶ For any other concept d in $\mathcal{G}$ which can be written as $p(d)0$ or $p(d)1$ such that exactly one of these alternatives is in $\mathcal{G}$, $p(d)$ belongs to $\mathcal{G}_1$.
- ▶ Thus the concepts in $\mathcal{G}$ are split into two groups, one consisting of strings which are all the same except for the last bit, and the other consisting of strings which have a unique prefix with respect to their projection onto S' .
- ▶ The arguments given above imply that $|\mathcal{G}_1| + |\mathcal{G}_2| = |\mathcal{G}|$.

Proof of Sauer's lemma



- ▶ $|\mathcal{G}_1|$ is the number of distinct concepts over a particular sample $x_1, x_2, \dots, x_{m-1}$.
- ▶ Since $VCdim(\mathcal{G}_1) \leq VCdim(\mathcal{G}) \leq d$, then by the definition of the growth function and using the induction hypothesis we have

$$|\mathcal{G}_1| \leq \Pi_{\mathcal{G}_1}(m-1) \leq \sum_{i=0}^{i=d} \binom{m-1}{i}$$

- ▶ By the definition of $\mathcal{G}_2$, if a set $Z \subset S'$ is shattered by $\mathcal{G}_2$, then $Z \cup \{x_m\}$ is shattered by $\mathcal{G}$. Therefore $VCdim(\mathcal{G}_2) \leq VCdim(\mathcal{G}) - 1 = d - 1$.

Proof of Sauer's lemma



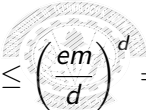
- ▶ By the definition of the growth function and the induction hypothesis, we have $|\mathcal{G}_2| \leq \Pi_{\mathcal{G}_2}(m-1) \leq \sum_{i=0}^{d-1} \binom{m-1}{i}$
- ▶ Thus we have

$$\begin{aligned} |\mathcal{G}| &= |\mathcal{G}_1| + |\mathcal{G}_2| \\ &\leq \sum_{i=0}^d \binom{m-1}{i} + \sum_{i=0}^{d-1} \binom{m-1}{i} \\ &= \sum_{i=0}^d \binom{m}{i} \end{aligned}$$

Corollary 3.18



Let $\mathcal{H}$ be a hypothesis set with $VCdim(\mathcal{H}) = d$. Then for all $m \geq d$,


$$\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d = O(m^d)$$

Proof of Corollary 3.18



$$\Pi_{\mathcal{H}}(m) = \sum_{i=1}^{i=d} \binom{m}{i}$$

$$\leq \sum_{i=1}^{i=d} \binom{m}{i} \left(\frac{m}{d}\right)^{d-i}$$

$$\leq \sum_{i=1}^{i=m} \binom{m}{i} \left(\frac{m}{d}\right)^{d-i}$$

$$\leq \left(\frac{m}{d}\right)^d \sum_{i=1}^{i=m} \binom{m}{i} \left(\frac{d}{m}\right)^i$$

$$\leq \left(\frac{m}{d}\right)^d \left(1 + \frac{d}{m}\right)^m \leq \left(\frac{m}{d}\right)^d e^d$$